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Department of Pre-University Education, Karnataka

II PUC · PHYSICS (33) · 2026–27

2 Mark, 3 Mark & 5 Mark Guide

Chapter-wise solved questions from the KSEAB Model Question Papers 1 – 5, written as per the Board's Scheme of Evaluation and the NCERT textbook

Department of Pre-University Education

Vijayapura

PART – B

TWO MARKS QUESTIONS

MARKS

CHAPTER 1

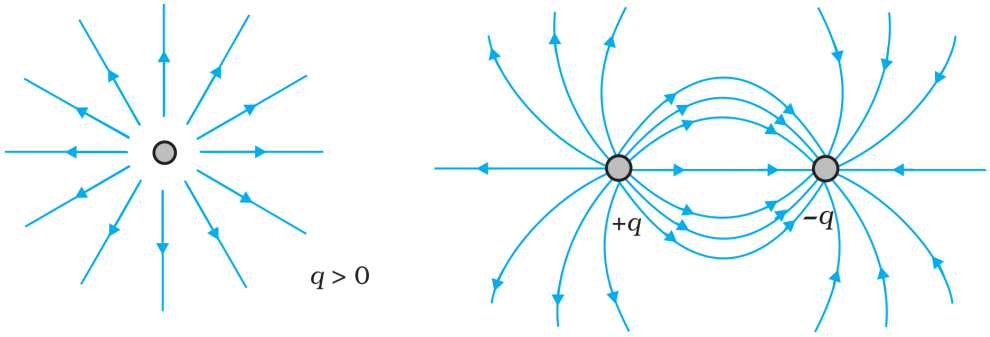
Electric Charges and Fields

1. Define electric field at a point. What is meant by 'source charge'? [MQP-1, Q.21]

Ans: Electric field at a point is the electrostatic force experienced by a unit positive (test) charge placed at that point. 1

The charge which produces the electric field is called the source charge. 1

2. Sketch the electric field lines for (a) a positive point charge and (b) an electric dipole. [MQP-2, Q.21]



(a) (d)

NCERT Fig. 1.14 — Field lines due to (a) a positive point charge and (d) an electric dipole (Physics Part I, Ch. 1)

(Arrows on the field lines are essential: outward for a positive charge; from $+q$ to $-q$ for a dipole.) 1+1

3. Define electric flux through an area element. Mention its SI unit. [MQP-3, Q.21]

Ans: Electric flux through an area element is the product of the electric field and the component of the area element perpendicular to the field (the number of field lines crossing the area). 1

SI unit: $\mathrm{N\,C^{-1}\,m^2}$ OR $\mathrm{V\,m}$ OR $\mathrm{J\,C^{-1}\,m}$ OR $\mathrm{N\,A^{-1}\,s^{-1}\,m^2}$ 1

OFFICIAL ANSWER SI unit as in KSEAB Scheme of Evaluation, II PUC Annual Exam March 2020, Q.1

4. Define the term 'linear charge density'. Mention its SI unit. [MQP-4, Q.21]

Ans: Linear charge density is the charge per unit length of a line charge. 1

SI unit: coulomb per metre ($\mathrm{C\,m^{-1}}$). 1

5. Mention two factors on which electric field at a point due to a point charge depends. [MQP-5, Q.21]

Ans: (i) Magnitude of the source charge q ($E \propto q$). 1

(ii) Distance r of the point from the charge ($E \propto 1/r^2$). 1

(iii) Permittivity of the medium. (Any two)

6. The electrostatic potential energy of a system of two like charges is positive. Explain. [MQP-1, Q.22]

Ans: Potential energy of two point charges: $U = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r}$. 1

For like charges $q_1 q_2 > 0$, so $U > 0$. Positive work has to be done against the force of repulsion to bring them closer from infinity; this work is stored as positive potential energy. 1

7. What are polar molecules? Give an example. [MQP-2, Q.22]

Ans: The molecules in which the centres of positive charges and negative charges do not coincide are called polar molecules. They have a permanent (intrinsic) dipole moment. 1

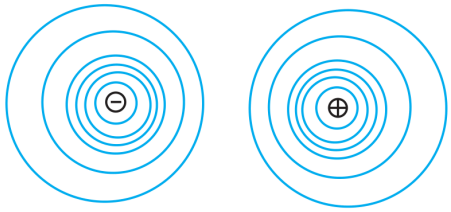
Example: water (H_2O), HCl. 1

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Supplementary Exam August 2021, Q.12 (definition)

8. What is meant by equipotential surface? Draw equipotential surfaces for an electric dipole. [MQP-3, Q.22]

Ans: An equipotential surface is a surface with a constant value of potential at all points on the surface. 1

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Annual Exam March 2018, Q.1



(a)

NCERT Fig. 2.11 (a) — Equipotential surfaces for a dipole (Physics Part I, Ch. 2)

1

9. Write two properties of equipotential surfaces. [MQP-4, Q.22]

Ans: (i) For any charge configuration, the equipotential surface through a point is normal to the electric field at that point.

(ii) Work done to move a charge on an equipotential surface is zero. 2

(iii) The equipotential surfaces corresponding to different potentials are (a) very close in case of a strong field and (b) far apart in case of a weak field.

(iv) Two equipotential surfaces do not intersect. (Any two)

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Supplementary Exam May 2023, Q.21

10. The electrostatic potential energy of a system of two like charges decreases with increase in the distance of separation. Explain the statement using suitable expression. [MQP-5, Q.22]

Ans: $U = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r}$ 1

For like charges $q_1 q_2$ is positive, so $U \propto \frac{1}{r}$. As the separation r increases, U decreases (the repulsive force does work as the charges move apart). 1

11. Name the mobile charge carriers in metals. Write the expression for mobility in terms of average collision time (relaxation time) for charge carriers. [MQP-1, Q.23]

Ans: Mobile charge carriers in metals: free electrons. 1

Mobility: $\mu = \frac{|v_d|}{E} = \frac{e\tau}{m}$, where τ = relaxation time, m = mass and e = charge of the electron. 1

12. State and explain Ohm's law. [MQP-2, Q.23]

Ans: **Ohm's law:** The electric current (I) flowing through a substance (conductor) is directly proportional to the voltage (potential difference, V) across its ends, provided the temperature and other physical conditions remain constant (same). 1

If I is the current through the conductor and V the potential difference between its ends, $V \propto I$, i.e. $V = IR$, where the constant R is the resistance of the conductor. 1

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Supplementary Exam August 2021, Q.21 and Annual Exam March 2017, Q.13

13. Write two limitations of Ohm's law. [MQP-3, Q.23]

Ans: (1) V ceases to be proportional to I. 1

(2) Semiconductor diodes do not obey Ohm's law. +

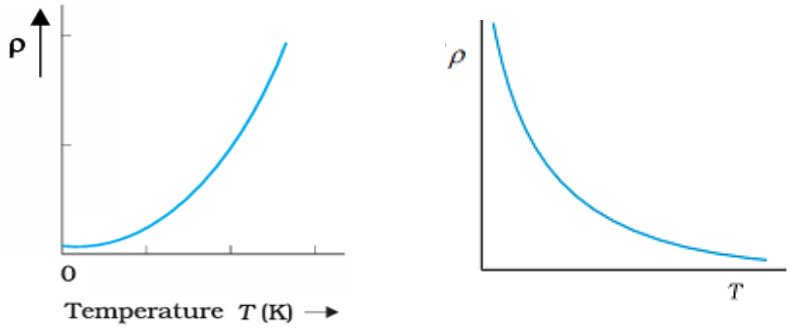
(3) For some materials like GaAs, the relation between V and I is not unique. 1

(4) Ohm's law is not applicable at very high and very low temperatures.

(Any two) (Any other correct limitation should be considered)

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Exam-1 March 2025, Q.22

14. Draw the graphs to show the variation of resistivity with temperature in case of (a) conductors and (b) semiconductors. [MQP-4, Q.23]



(a) Conductor (copper) *(b) Semiconductor (silicon)*

1+1

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Exam-2 2026, Q.23 (graphs reproduced from the scheme)

(For conductors resistivity increases with temperature; for semiconductors it decreases.)

15. State and explain Kirchhoff's loop rule. [MQP-5, Q.23]

Ans: **Loop rule:** In an electrical network, the algebraic sum of changes in potential around any closed loop involving resistors and cells in the loop is zero. 1

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Exam-2 April/May 2025, Q.23

Explanation: Electric potential depends only on position, so on returning to the starting point the total change in potential is zero ($\sum \Delta V = 0$). The rule is based on the law of conservation of energy. E.g. for a loop with a cell ε and resistors R_1, R_2: $\varepsilon - IR_1 - IR_2 = 0$. 1

16. What is the nature of the force between two parallel conductors carrying currents in the (a) same direction and (b) opposite directions? [MQP-1, Q.24]

Ans: (a) Same direction: **attractive**. 1

(b) Opposite directions: **repulsive**. 1

17. A current loop of magnetic dipole moment 10 A m^2 is suspended freely in a uniform magnetic field of magnitude 5×10^{-3} T. The plane of the coil is in the direction of the magnetic field. Calculate the torque on the current loop. [MQP-2, Q.24]

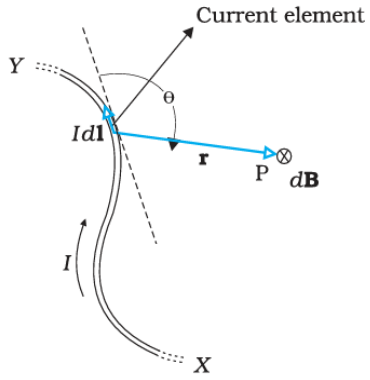
Ans: Torque: $\tau = mB \sin\theta$. Since the plane of the coil is along $\vec{B}$, $\vec{m} \perp \vec{B}$, so $\theta = 90°$. 1

$\tau = 10 \times 5 \times 10^{-3} \times \sin 90° \Rightarrow \boxed{\tau = 5 \times 10^{-2}\ \text{N m}}$ 1

18. Give the vector form of Biot–Savart law and explain the terms. [MQP-3, Q.24]

Ans: $d\vec{B} = \dfrac{\mu_0}{4\pi} \dfrac{I\, d\vec{l} \times \vec{r}}{r^3}$ OR $d\vec{B} = \dfrac{\mu_0}{4\pi} \dfrac{I\, d\vec{l} \times \hat{r}}{r^2}$ 1

$d\vec{B}$ = magnetic field at the point due to the current element $I\, d\vec{l}$; $\vec{r}$ = position vector of the point from the element ($\hat{r}$ its unit vector); μ_0 = permeability of free space ($4\pi \times 10^{-7}$ T m A^{-1}). 1



Current element $I\, d\vec{l}$ and the field $d\vec{B}$ at P (from the KSEAB scheme)

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Exam-2 2026, Q.31 (expression and diagram)

19. A straight wire of length 0.5 m carrying a current of 2 A is placed in a uniform magnetic field of 0.3 T. Calculate the maximum magnetic force on the wire. [MQP-4, Q.24]

Ans: $F = BIl \sin\theta$; it is maximum when $\theta = 90°$: $F_{max} = BIl$ 1

$F_{max} = 0.3 \times 2 \times 0.5 \Rightarrow \boxed{F_{max} = 0.3\ \text{N}}$ 1

20. When is the force on a charged particle moving in a uniform magnetic field (a) maximum and (b) zero? [MQP-5, Q.24]

The magnitude of the force is $F = qvB \sin\theta$, where θ is the angle between $\vec{v}$ and $\vec{B}$ (KSEAB Supplementary Exam May 2023, Q.22).

Ans: (a) **Maximum** ($F = qvB$) when the particle moves perpendicular to the field ($\theta = 90°$). 1

(b) **Zero** when the particle moves parallel or antiparallel to the field ($\theta = 0°$ or $180°$). 1

CHAPTER 6

Electromagnetic Induction

21. Calculate the magnetic energy stored in a coil of self-inductance 0.05 H carrying a current of 0.2 A. [MQP-1, Q.25]

Ans: $U = \frac{1}{2}LI^2$ 1

$U = \frac{1}{2} \times 0.05 \times (0.2)^2 \Rightarrow \boxed{U = 1 \times 10^{-3}\,\text{J}}$ 1

22. Mention two factors on which self-inductance of a solenoid depends. [MQP-2, Q.25]

Ans: (i) Number of turns per unit length n (ii) Area of cross-section A 1

(iii) Length l of the solenoid (iv) Relative permeability μ_r of the core material (Any two) 1

23. What is AC generator? What is its working principle? [MQP-3, Q.25]

Ans: An AC generator is a device which converts mechanical energy into electrical energy (alternating emf). 1

Principle: Electromagnetic induction. When a coil is rotated in a magnetic field, the magnetic flux linked with it changes and an emf is induced. 1

24. State and explain Faraday's law of electromagnetic induction. [MQP-4, Q.25]

Ans: The magnitude of the induced emf in a circuit is equal to the time rate of change of magnetic flux through the circuit. **OR** The magnitude of the induced emf in a circuit is directly proportional to the rate of change of magnetic flux linked with the circuit. 1

If $d\phi_B$ is the change in magnetic flux in time dt, then the induced emf is $\varepsilon = -\dfrac{d\phi_B}{dt}$. 1

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Annual Exam March 2017, Q.15

25. Give the expression for mutual inductance between two co-axial solenoids and explain the terms. [MQP-5, Q.25]

Ans: $M = \mu_0 n_1 n_2 \pi r_1^2 l$ 1

n_1, n_2 = number of turns per unit length of the inner and outer solenoids, r_1 = radius of the inner solenoid, l = length of the solenoids, μ_0 = permeability of free space. 1

CHAPTER 8

Electromagnetic Waves

26. Name the electromagnetic waves used in: (a) remote switches of TV sets, (b) the laser assisted eye surgery (LASIK). [MQP-1, Q.26]

Ans: (a) Remote switches of TV sets → **Infra-red rays** 1

(b) LASIK surgery → **Ultraviolet rays** 1

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Supplementary Exam September 2020, Q.17 and Exam-1 March 2026, Q.26

27. What is displacement current? Write the mathematical form of Ampere–Maxwell law. [MQP-2, Q.26]

Ans: The current due to changing electric flux (or electric field) with time is called displacement current. 1

Ampere–Maxwell law: $\oint \vec{B} \cdot d\vec{l} = \mu_0 i_c + \mu_0 \varepsilon_0 \dfrac{d\phi_E}{dt}$ 1

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Exam-2 2026, Q.26

28. List two uses of microwaves. [MQP-3, Q.26]

Ans: (1) They are used in radar systems used in aircraft navigation. 1

(2) They are used in speed guns. +

(3) They are used in microwave ovens. 1

(Any two) (Any other relevant use should be considered)

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Exam-1 March 2025, Q.25

29. Name the electromagnetic waves generated by: (a) the accelerated motion of charges in conducting wires, (b) the bombardment of a metal target by high energy electrons. [MQP-4, Q.26]

Ans: (a) Radio waves 1

(b) X-rays 1

30. Mention two applications of X-rays. [MQP-5, Q.26]

Ans: (a) X-rays are used as a diagnostic tool in medicine. 1

(b) To treat certain forms of cancer. 1

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Annual Exam March 2018, Q.15

CHAPTER 12
Atoms

31. Give two limitations of Bohr's atomic model. [MQP-1, Q.27]

Ans: (i) Bohr's model is applicable only to hydrogenic (single electron) atoms. **OR** Bohr's model is not applicable to atoms with two or more electrons. 1

(ii) This model is unable to explain the relative intensities of the frequencies emitted by hydrogenic (hydrogen-like) atoms. (Any other correct answer) 1

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Exam-2 2026, Q.27

32. Define impact parameter. What is the value of angle of scattering for minimum impact parameter? [MQP-2, Q.27]

Ans: The impact parameter is the perpendicular distance of the initial velocity vector of the α-particle from the centre of the target nucleus. 1

Angle of scattering is 180° (for zero, i.e. minimum, impact parameter the α-particle rebounds). 1

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Exam-1 March 2026, Q.27

33. The radius of the innermost electron orbit of a hydrogen atom is 5.3×10^{-11} m. Determine the radius of the $n = 2$ orbit. [MQP-3, Q.27]

Ans: $r_n = n^2 r_1$ 1

$r_2 = 2^2 \times 5.3 \times 10^{-11} \Rightarrow \boxed{r_2 = 2.12 \times 10^{-10}\ \text{m}}$ 1

34. 'Stable structure of an atom as given by Rutherford model is not possible according to classical electromagnetic theory'. Explain this statement. [MQP-4, Q.27]

Ans: According to classical electromagnetic theory, an accelerated charge radiates energy. An electron revolving in a circular orbit is accelerated (centripetal acceleration), so it continuously loses energy. 1

Its orbit would shrink and the electron would spiral into the nucleus. Hence the atom cannot be stable (and it would emit a continuous spectrum, contrary to the observed line spectra). 1

35. Calculate the angular momentum of an electron revolving in the 1st orbit of hydrogen atom. [MQP-5, Q.27]

Ans: $L = \frac{nh}{2\pi}$; for $n = 1$, $L = \frac{h}{2\pi}$ 1

$L = \frac{6.63 \times 10^{-34}}{2 \times 3.14} \Rightarrow \boxed{L = 1.055 \times 10^{-34}\ \text{J s}}$ 1

CHAPTER 14
Semiconductor Electronics

36. List out two differences between p-type and n-type semiconductors. [MQP-1, Q.28]

Ans:

p-type semiconductors	n-type semiconductors
1. They are obtained by doping pure semiconductors with trivalent impurity.	1. They are obtained by doping pure semiconductors with pentavalent impurity.
2. Holes are the majority charge carriers.	2. Electrons are the majority charge carriers.
3. Electrons are the minority charge carriers.	3. Holes are the minority charge carriers.

1+1

(Any two; any other correct difference must be considered)

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Exam-1 March 2026, Q.28

37. Give any two differences between intrinsic and extrinsic semiconductors. [MQP-2, Q.28]

Ans:

Intrinsic semiconductors	Extrinsic semiconductors
Semiconductor without doping (pure form).	Semiconductor doped with impurity atoms (impure form).
Number of free electrons (n_e) is equal to the number of holes (n_h), $n_e = n_h$.	Number of free electrons and holes are unequal, $n_e \neq n_h$.
Conductivity is low.	Conductivity is high.
Conductivity depends only on temperature.	Conductivity depends both on temperature and impurity added.
Conductivity is equally due to electrons and holes.	Conductivity is mainly due to majority charge carriers.

1+1

(Any two — any other correct difference)

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Exam-2 2026, Q.28

38. Mention two differences between half wave rectifier and full wave rectifier. [MQP-3, Q.28]

Ans:

Half-wave rectifier	Full-wave rectifier
Rectifies only one half cycle of the AC input.	Rectifies both half cycles of the AC input.
Uses one diode.	Uses two diodes (with a centre-tap transformer).
Efficiency is low (about 40.6%).	Efficiency is high (about 81.2%).
Output has more ripple.	Output has less ripple.

1+1

(Any two)

39. What are dopants? Name the type of dopants used to obtain p-type semiconductors. [MQP-4, Q.28]

Ans: Dopants are the impurity atoms deliberately added to a pure semiconductor to increase its conductivity. 1

p-type: **trivalent** (acceptor) impurities, e.g. boron, aluminium, indium, gallium. 1

40. What is meant by energy gap? Give its value for a conductor. [MQP-5, Q.28]

Ans: The energy difference between the bottom of the conduction band and the top of the valence band is called the energy gap (forbidden gap). No electron energy states exist in this gap. 1

For a conductor, the energy gap is **zero** (the valence and conduction bands overlap). 1

OFFICIAL ANSWER Value for a conductor: KSEAB Scheme of Evaluation, II PUC Supplementary Exam May 2023, Q.20

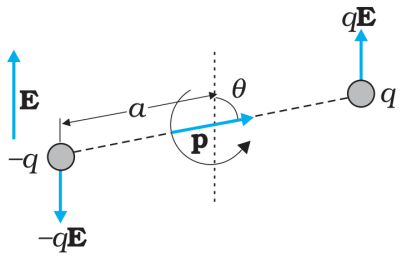
PART – C
THREE MARKS QUESTIONS

MARKS

CHAPTER 1
Electric Charges and Fields

1. Derive an expression for torque acting on an electric dipole placed in a uniform electric field. [MQP-1, Q.29]

Ans: Consider an electric dipole of dipole moment $\vec{p}$ placed in a uniform external electric field $\vec{E}$, making an angle θ with it. Force on $+q$ is $q\vec{E}$ and on $-q$ is $-q\vec{E}$; the net force is zero but they form a couple. 1



NCERT Fig. 1.19 — Dipole in a uniform electric field (Physics Part I, Ch. 1)

Magnitude of torque = force × perpendicular distance (force on q is $F = qE$) 1

$\tau = qE \times (2a \sin\theta) = (2aq)E \sin\theta \quad (\because\ 2aq = p$, dipole moment)

$\boxed{\tau = pE \sin\theta}$ or in vector form $\vec{\tau} = \vec{p} \times \vec{E}$ 1

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Exam-2 2026, Q.29

2. Give Coulomb's law in vector form and explain the terms. Define SI unit of charge using Coulomb's law. [MQP-2, Q.29]

Ans: Electrostatic force on charge q_2 by charge q_1 is: $\vec{F}_{21} = \dfrac{1}{4\pi\varepsilon_0}\dfrac{q_1 q_2}{r_{21}^2}\hat{r}_{21}$ 1

where q_1, q_2 are the charges, r_{21} is the distance between the charges, $\hat{r}_{21}$ is the unit vector in the direction from q_1 towards q_2. 1

One coulomb is the charge on two identical point charges which are separated by a distance of 1 m in vacuum and experience an electric force of repulsion of 9×10^9 N. 1

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Exam-1 March 2026, Q.29

3. Define linear charge density. Mention the expression for electric field at a point due to an infinitely long uniformly charged wire and explain the terms. [MQP-3, Q.29]

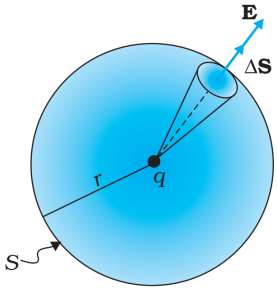
Ans: Linear charge density is the charge per unit length of the wire (SI unit C m^{-1}). 1

$E = \dfrac{\lambda}{2\pi\varepsilon_0 r}$ 1

λ = linear charge density of the wire, r = perpendicular distance of the point from the wire, ε_0 = permittivity of free space. The field is radial (outward for $\lambda > 0$). 1

4. Show that the total electric flux through the surface of a sphere enclosing a point charge 'q' at its centre is $\frac{q}{\varepsilon_0}$. [MQP-4, Q.29]

Ans: Consider a sphere of radius r with a point charge q at its centre. The field at every point on the surface is radial with magnitude $E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$. 1



NCERT Fig. 1.22 — Flux through a sphere enclosing a point charge q at its centre (Physics Part I, Ch. 1)

For an area element $\Delta\vec{S}$, $\vec{E}$ and $\Delta\vec{S}$ are along the same direction: $\Delta\phi = \vec{E}\cdot\Delta\vec{S} = E\,\Delta S$ 1

Total flux: $\phi = \sum E\,\Delta S = E\sum\Delta S = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\times 4\pi r^2 \Rightarrow \boxed{\phi = \frac{q}{\varepsilon_0}}$ 1

5. Mention three properties of electric field lines. [MQP-5, Q.29]

Ans: (1) Field lines start from positive charges and end at negative charges. 1

(2) In a charge-free region, electric field lines can be taken to be continuous curves without any breaks. +

(3) Two field lines can never cross each other. 1

(4) Electrostatic field lines do not form any closed loops. +

(5) A tangent drawn to an electric field line at any point gives the direction of the electric field at that point. 1

(6) Density of field lines is a measure of the strength of the electric field. (Any three)

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Exam-1 March 2025, Q.29

CHAPTER 2

Electrostatic Potential and Capacitance

6. What is a capacitor? Mention two factors on which capacitance of a parallel plate capacitor depends. [MQP-1, Q.30]

Ans: A capacitor is a device consisting of a pair of conductors separated by an insulator, which can be used to store charges. 1

Capacitance of a capacitor depends on: (i) area of the plates; (ii) distance between the plates; (iii) medium between the plates (or dielectric constant of the medium between the plates). (Any two factors) 1+1

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Exam-1 March 2026, Q.30

7. Mention any three important results regarding electrostatics of conductors. [MQP-2, Q.30]

Ans: (a) The interior of a conductor can have no excess charge in the static situation. 1

(b) Inside a conductor, electrostatic field is zero. 1

(c) On the surface of a charged conductor, electrostatic field must be normal to the surface at every point. 1

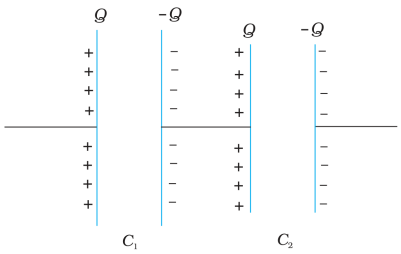
(d) Electrostatic potential is constant throughout the volume of the conductor and has the same value (as inside) on its surface.

(e) Electric field at the surface of a charged conductor, $\vec{E} = \frac{\sigma}{\varepsilon_0}\hat{n}$, where σ is the surface charge density and $\hat{n}$ is a unit vector normal to the surface. (Any three)

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Exam-2 2026, Q.30

8. Obtain the expression for equivalent capacitance of two capacitors connected in series. [MQP-3, Q.30]

Ans: Let two capacitors with capacitances C_1 and C_2 be joined in series. In the series combination, the charges on the two plates ($\pm Q$) are the same on each capacitor. The total potential drop V across the combination is the sum of the potential drops V_1 and V_2 across C_1 and C_2.



NCERT Fig. 2.26 — Combination of two capacitors in series (Physics Part I, Ch. 2)

Then $V = V_1 + V_2 = \frac{Q}{C_1} + \frac{Q}{C_2} = Q\left(\frac{1}{C_1} + \frac{1}{C_2}\right)$ .....(1) 1

(since $Q = C_1V_1 = C_2V_2 \Rightarrow V_1 = Q/C_1$ and $V_2 = Q/C_2$)

The combination has an effective capacitance C_s with charge Q and potential difference V: 1

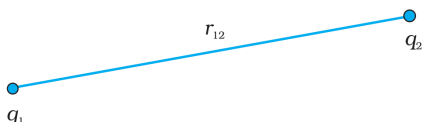
$C_s = \frac{Q}{V} \Rightarrow V = \frac{Q}{C_s}$ ..... (2)

Comparing (2) with (1): $\frac{Q}{C_s} = Q\left(\frac{1}{C_1} + \frac{1}{C_2}\right) \Rightarrow \boxed{\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2}}$ 1

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Supplementary Exam August 2021, Q.20

9. Derive an expression for the potential energy of a system of two point charges in the absence of external electric field. [MQP-4, Q.30]

Ans: Consider charges q_1 and q_2 brought from infinity to points $\vec{r}_1$ and $\vec{r}_2$. First q_1 is brought: there is no field, so the work done is zero. 1

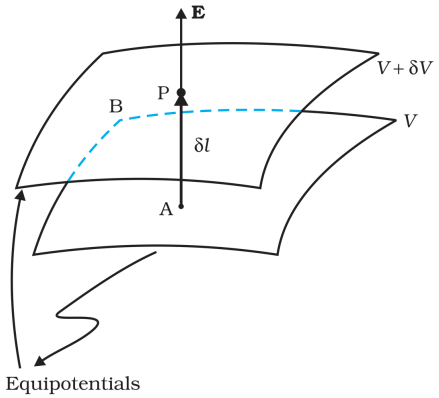


NCERT Fig. 2.13 — Potential energy of a system of charges q_1 and q_2 (Physics Part I, Ch. 2)

Next q_2 is brought against the field of q_1. Work done = $q_2\times$ potential of q_1 at $\vec{r}_2$: $W = q_2 \cdot \frac{1}{4\pi\varepsilon_0}\frac{q_1}{r_{12}}$ 1

This work is stored as potential energy: $\boxed{U = \frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{r_{12}}}$, where r_{12} is the distance between the charges. 1

10. Derive the relation between electric field and electric potential. [MQP-5, Q.30]



NCERT Fig. 2.12 — From the potential to the field (Physics Part I, Ch. 2)

Ans: Consider two closely spaced equipotential surfaces A and B with potentials V and $V + \delta V$. Work done to move a unit positive charge from P to A (distance δl, against the field) is $W = F\,\delta l = E\,\delta l$ (for unit positive charge, $q = 1$ C, force $F = E$). 1

This work done is equal to the potential difference: $W = q(V_A - V_B) = 1 \cdot [V - (V + \delta V)] = -\delta V$ 1

Thus $E\,\delta l = -\delta V \quad \therefore \quad \boxed{E = -\dfrac{\delta V}{\delta l}}$ (The field is in the direction in which the potential decreases steepest.) 1

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Supplementary Exam September 2020, Q.19

CHAPTER 4
Moving Charges and Magnetism

11. In a chamber, a uniform magnetic field of 6.5×10^{-4} T is maintained. An electron is shot into the field with a speed of 4.8×10^{6} m s^{-1} normal to the field. Determine the radius of the circular orbit. Given: $e = 1.6 \times 10^{-19}$ C and $m_e = 9.1 \times 10^{-31}$ kg. [MQP-1, Q.31]

Ans: The magnetic force provides the centripetal force: $qvB = \dfrac{mv^2}{r} \Rightarrow r = \dfrac{mv}{qB}$ 1

$$r = \frac{9.1 \times 10^{-31} \times 4.8 \times 10^{6}}{1.6 \times 10^{-19} \times 6.5 \times 10^{-4}}$$ 1

$\boxed{r = 4.2 \times 10^{-2} \text{ m} = 4.2 \text{ cm}}$ 1

12. What is Lorentz force? Write the expression representing this force and explain the terms. [MQP-2, Q.31]

Ans: The total force on a moving charge in the presence of both electric and magnetic fields is called the Lorentz force. 1

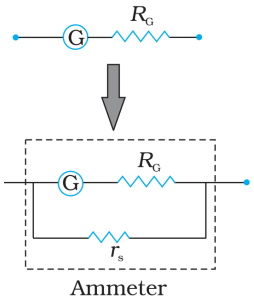
Expression: $\vec{F} = q\left[\vec{E} + (\vec{v} \times \vec{B})\right]$ 1

where q is the charge, $\vec{E}$ is the electric field, $\vec{B}$ is the magnetic field and $\vec{v}$ is the velocity of the charge. 1

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Exam-3 2025, Q.23 and Exam-1 March 2025, Q.23

13. Explain the conversion of galvanometer into an ammeter using relevant circuit diagram and expression. [MQP-3, Q.31]

Ans: A galvanometer can be converted into an ammeter by connecting a **low resistance in parallel** with it. 1



NCERT Fig. 4.21 — Conversion of a galvanometer (G) to an ammeter (Physics Part I, Ch. 4)

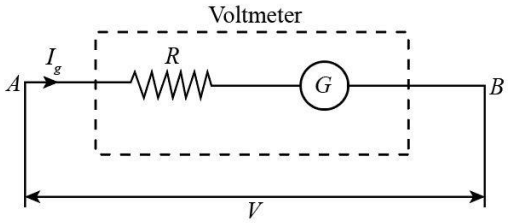
Circuit diagram (as above). R_G – resistance of galvanometer G; r_s – shunt resistance in parallel with the galvanometer. 1

The resistance of the arrangement = $\frac{R_G r_s}{R_G + r_s}$ OR shunt resistance $r_s = \frac{I_G R_G}{I - I_G}$ OR $S = \frac{I_g G}{I - I_g}$ 1

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Supplementary Exam May 2023, Q.32

14. Explain the conversion of galvanometer into a voltmeter using relevant circuit diagram and expression. [MQP-4, Q.31]

Ans: A galvanometer can be converted into a voltmeter by connecting a suitable **high resistance in series** with it. The circuit diagram is as shown below: 1



*Voltmeter: galvanometer G with a high resistance **R** in series (from the KSEAB scheme)*

1

The value of the series resistance is given by: $R = \frac{V}{I_g} - G$ 1

where V = range of the voltmeter, I_g = maximum current through the galvanometer and G = resistance of the galvanometer. (Other symbols should be considered)

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Exam-1 March 2026, Q.31

15. Define 'current sensitivity' of a galvanometer. Write two factors on which current sensitivity depends. [MQP-5, Q.31]

Ans: Current sensitivity of a galvanometer is the deflection produced per unit current through it. 1

Current sensitivity of a galvanometer depends on: * number of turns in the coil (N); * area of the coil (A); * magnetic field used (B); * torsional constant of the spring (k). 1+1 (any two)

OFFICIAL ANSWER Factors: KSEAB Scheme of Evaluation, II PUC Exam-2 April/May 2025, Q.24

16. Mention three properties of magnetic field lines. [MQP-1, Q.32]

Ans: (i) The magnetic field lines form closed loops.

(ii) The tangent to the field line at a given point represents the direction of the net magnetic field at that point.

(iii) The larger the number of field lines crossing per unit area, the stronger is the magnitude of the magnetic field. 3

(iv) The magnetic field lines do not intersect. (Any 3 properties)

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Supplementary Exam May 2023, Q.33

17. State and explain Gauss's law in magnetism. Give its significance. [MQP-2, Q.32]

Ans: **Statement:** The net magnetic flux through any closed surface placed in a magnetic field is zero. 1

Explanation: The magnetic field lines always form a closed loop. Therefore the total magnetic flux through the closed surface $\Phi = \sum \vec{B} \cdot \Delta\vec{S} = 0$. **OR** The isolated magnetic poles do not exist; therefore $\Phi = \sum \vec{B} \cdot \Delta\vec{S} = 0$. 1

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Supplementary Exam May 2023, Q.23

Significance: Isolated magnetic poles (magnetic monopoles) are not known to exist; magnetic field lines form closed loops. 1

18. Write three differences between diamagnetic and paramagnetic materials. [MQP-3, Q.32]

Ans:

Diamagnetic substances	**Paramagnetic substances**
1. These are repelled by a magnet.	These are attracted by a magnet.
2. The magnetic susceptibility is negative, $\chi < 0$.	The magnetic susceptibility is positive, $\chi > 0$.
3. The susceptibility (or permeability or magnetisation) does not depend on temperature. OR They do not obey Curie's law.	The susceptibility (or permeability or magnetisation) depends on temperature. OR They obey Curie's law.
4. Magnetic field lines are expelled out when the substance is placed in an external magnetic field.	Magnetic field lines enter inside when the substance is placed in an external magnetic field.
5. Relative permeability is less than one, $\mu_r < 1$.	Relative permeability is more than one, $\mu_r > 1$.

1+1+1

(Any three of these OR any other correct difference)

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Annual Exam March 2023, Q.33

19. What are paramagnetic substances? Mention two factors on which susceptibility of a paramagnetic material depends. [MQP-4, Q.32]

Ans: Paramagnetic materials magnetise weakly in an external magnetic field in the direction of the magnetising field; they are weakly attracted by a powerful magnet. E.g. aluminium, sodium, calcium, oxygen. 1

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Exam-1 March 2024, Q.34

(i) Absolute temperature: the susceptibility of a paramagnetic substance varies inversely with its absolute temperature, $\chi \propto \frac{1}{T}$ (Curie's law). 1

(ii) Nature of the material (Curie constant C). 1

20. Obtain the expression for the potential energy of a magnetic needle pivoted in uniform magnetic field. [MQP-5, Q.32]

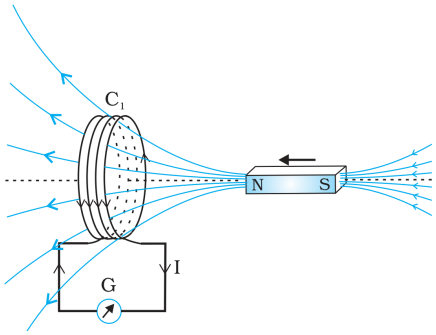
Ans: A needle of magnetic moment $\vec{m}$ at angle θ to $\vec{B}$ experiences a torque $\tau = mB\sin\theta$. 1

Work done in rotating it through $d\theta$: $dW = \tau\, d\theta$. Hence $U = \int mB\sin\theta\, d\theta = -mB\cos\theta$ (taking $U = 0$ at $\theta = 90°$). 1

$\boxed{U = -\vec{m}\cdot\vec{B} = -mB\cos\theta}$. U is minimum ($-mB$) at $\theta = 0°$ (stable) and maximum ($+mB$) at $\theta = 180°$ (unstable). 1

CHAPTER 6
Electromagnetic Induction

21. Describe the coil and bar magnet experiment to demonstrate the phenomenon of electromagnetic induction. [MQP-1, Q.33]



NCERT Fig. 6.1 — When the bar magnet is pushed towards the coil, the galvanometer G deflects (Physics Part I, Ch. 6)

Ans: Diagram (as above). 1

• When the North-pole of a bar magnet is pushed towards the coil, the pointer in the galvanometer deflects. 1

• The galvanometer does not show any deflection when the magnet is held stationary. 1

• When the magnet is pulled away from the coil, the galvanometer shows deflection in the opposite direction, which indicates reversal of the current's direction.

• When the South-pole of the bar magnet is moved towards or away from the coil, the deflections in the galvanometer are opposite to that observed with the North-pole for similar movements.

• The deflection is found to be larger when the magnet is pushed towards or pulled away from the coil faster.

• When the bar magnet is held fixed and the coil C_1 is moved towards/away from the magnet, same effects are observed.

• The relative motion between the magnet and the coil is responsible for generation (induction) of electric current in the coil. (ANY TWO OBSERVATIONS)

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Exam-2 2026, Q.33

22. Derive the expression for the magnetic energy stored in a coil, $U = \frac{1}{2}LI^2$. [MQP-2, Q.33]

Ans: During setting-up a current of I in a coil of self-inductance L, the instantaneous power is: 1

$$p = |\varepsilon|\, i = Li\frac{di}{dt}$$

Therefore, the magnetic energy stored in the coil (which is equal to energy spent in the process) is: 1

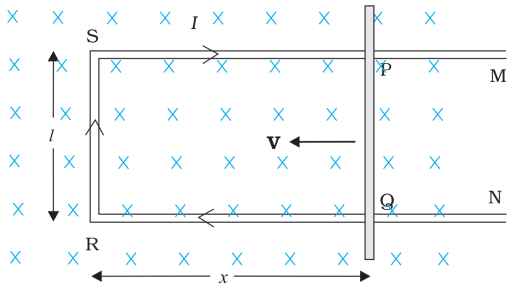
$$U = \int p\, dt = \int_0^I Li\frac{di}{dt}\, dt$$

$$\Rightarrow \boxed{U = \tfrac{1}{2}LI^2}$$ 1

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Exam-1 March 2026, Q.33

23. Derive the expression for the motional emf induced in a conducting rod moving perpendicular to a uniform magnetic field. [MQP-3, Q.33]

Ans: Consider a rod PQ of length l moving perpendicular to a uniform magnetic field B with a speed v as shown in the figure.



NCERT Fig. 6.10 — The arm PQ is moved to the left side (Physics Part I, Ch. 6)

1

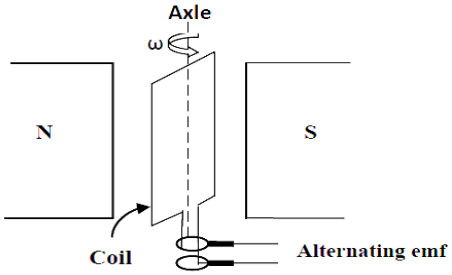
Magnetic flux for the surface PQRS: $\phi_B = BA\cos\theta = Blx$

Induced emf in the rod: $\varepsilon = -\dfrac{d\phi_B}{dt}$ 1

$\varepsilon = -\dfrac{d(Blx)}{dt} = \boxed{Blv}$ $\quad$ (since $-\dfrac{dx}{dt} = v$) 1

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Exam-1 March 2025, Q.33

24. Derive the expression for the emf generated in the coil of an AC generator. [MQP-4, Q.33]



AC generator: coil rotating between the poles N and S (from the KSEAB scheme)

Ans: Labelled diagram (as above). Let the coil be rotated with a constant angular speed ω; the angle θ between the magnetic field vector $\vec{B}$ and the area vector $\vec{A}$ of the coil at any instant t is $\theta = \omega t$. 1

Due to rotation, the effective area of the coil exposed to the magnetic field lines changes with time. 1

The magnetic flux at any time t is $\Phi_B = \vec{B}\cdot\vec{A} = BA\cos\theta = BA\cos\omega t$

From Faraday's law, the induced emf in the rotating coil of N turns: 1

$\varepsilon = -N\dfrac{d\Phi_B}{dt} = -NBA\dfrac{d}{dt}(\cos\omega t)$

$\varepsilon = NBA\omega\sin\omega t$, where $NBA\omega = \varepsilon_0$ is the maximum value of the emf. Thus the instantaneous emf is $\boxed{\varepsilon = \varepsilon_0\sin\omega t}$.

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Supplementary Exam August 2021, Q.29

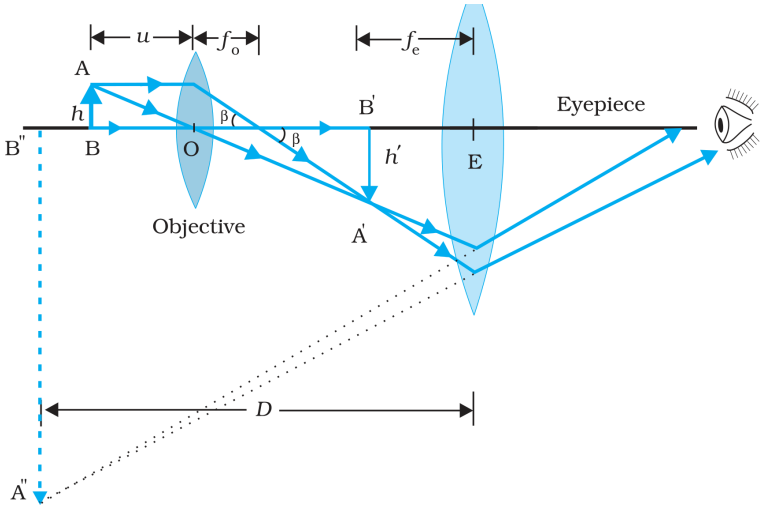
25. The pedals of a bicycle are attached to a 100-turn coil of area 0.10 m^2. The coil is rotated at half a revolution per second, and it is placed in a uniform magnetic field of 0.01 T perpendicular to the axis of rotation of the coil. Calculate the maximum voltage generated in the coil. [MQP-5, Q.33]

Ans: Maximum emf: $\varepsilon_0 = NBA\omega$ 1

$\omega = 2\pi\nu = 2\pi\times 0.5 = \pi\ \text{rad s}^{-1}$; $\varepsilon_0 = 100\times 0.01\times 0.10\times 3.14$ 1

$\boxed{\varepsilon_0 = 0.314\ \text{V}}$ 1

26. Draw the ray diagram for the formation of image by a compound microscope. Write the expression for linear magnification produced by the objective of compound microscope in terms of its tube length. [MQP-1, Q.34]



NCERT Fig. 9.24 — Ray diagram for the formation of image by a compound microscope (Physics Part II, Ch. 9)

2

Linear magnification by the objective: $m_o = \frac{L}{f_o}$, where L = tube length (distance between the second focal point of the objective and the first focal point of the eyepiece) and f_o = focal length of the objective. 1

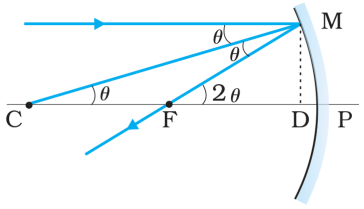
27. Give the Cartesian sign convention used to measure distances and heights in spherical mirrors. [MQP-2, Q.34]

Ans:
- All the distances are measured from the pole of the mirror. 1
- The distances measured in the same direction as the incident light are taken as positive and the distances measured in the direction opposite to the direction of incident light are taken as negative. 1
- The heights measured upwards (normal to the principal axis of the mirror) are taken as positive. 1

The heights measured downwards are taken as negative.

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Exam-2 2026, Q.34

28. Show that the focal length of spherical mirror is equal to half of its radius of curvature. [MQP-3, Q.34]



(a)

NCERT Fig. 9.4 (a) — Geometry of reflection of an incident ray on a concave spherical mirror (Physics Part II, Ch. 9)

Ans: Labelled diagram with arrows. $\angle MCP = \theta$ and $\angle MFP = 2\theta$. 1

$\tan\theta = \frac{MD}{CD}$ and $\tan 2\theta = \frac{MD}{FD}$.

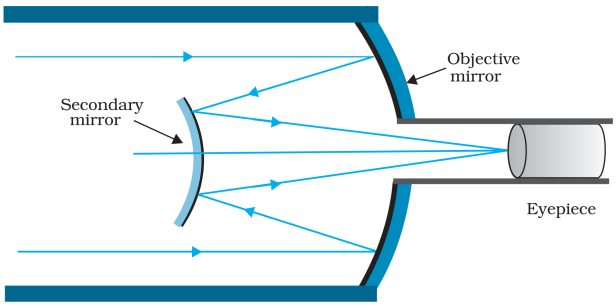
For small angle θ, $\tan\theta \approx \theta$ and $\tan 2\theta \approx 2\theta$. $\therefore\ 2\theta = \dfrac{MD}{FD},\ 2\dfrac{MD}{CD} = \dfrac{MD}{FD} \Rightarrow FD = \dfrac{CD}{2}$ 1

For small θ, the point D is very close to the point P. Therefore $FD = -f$ and $CD = -R$. Focal length: $\boxed{f = \dfrac{R}{2}}$ 1

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Annual Exam March 2023, Q.35

29. What are reflecting telescopes? Explain the advantages of mirror objectives used in them. [MQP-4, Q.34]

Ans: Telescopes which use a concave mirror (instead of a lens) as the objective are called reflecting telescopes, e.g. the Cassegrain telescope. 1



NCERT Fig. 9.26 — Schematic diagram of a reflecting telescope (Cassegrain) (Physics Part II, Ch. 9)

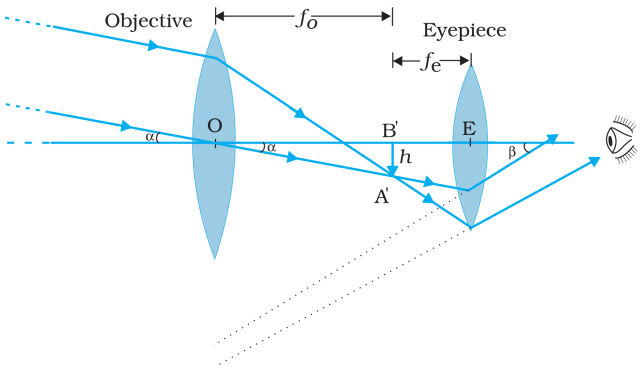
• There is no chromatic aberration in a mirror. 1

• Using a parabolic reflecting surface (mirror), spherical aberration is also removed. 1

• Providing mechanical support is much less of a problem, since a mirror weighs much less than a lens. (any two)

OFFICIAL ANSWER Advantages: KSEAB Scheme of Evaluation, II PUC Exam-3 2025, Q.27

30. Draw the ray diagram for the formation of image in case of a refracting telescope. Mention any one advantage of using objective of larger diameter in a refracting telescope. [MQP-5, Q.34]



NCERT Fig. 9.25 — A refracting telescope (Physics Part II, Ch. 9)

2

Advantage of a larger objective: greater light-gathering power (fainter objects can be seen) and higher resolving power. 1

31. What are matter waves? Explain de Broglie hypothesis of matter waves. [MQP-1, Q.35]

Ans: Waves associated with moving material particles are called matter waves (de Broglie waves). 1

de Broglie hypothesis: Nature is symmetrical; just as radiation has a dual nature, moving material particles should also display wave-like properties. 1

The wavelength associated with a particle of momentum p: $\lambda = \frac{h}{p} = \frac{h}{mv}$, where h is Planck's constant. 1

32. Write three properties of photon. [MQP-2, Q.35]

Ans: (a) In interaction of radiation with matter, radiation behaves as if it is made up of particles called photons.

(b) Each photon has energy $E = h\nu$, where ν is the frequency, and momentum $p = h\nu/c$, where c is the speed of light.

(c) All photons of light of a particular frequency (ν), or wavelength (λ), have the same energy and momentum. The photon energy is independent of intensity of radiation. 1+1+1

(d) Photons are electrically neutral and are not deflected by electric and magnetic fields.

(e) In a photon-particle collision (such as photon-electron collision), the total energy and total momentum are conserved. However, the number of photons may not be conserved in a collision.

(Any three)

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Supplementary Exam May 2023, Q.43(ii)

33. The work function of caesium is 2.14 eV. Calculate the threshold frequency for caesium. [MQP-3, Q.35]

Ans: $\phi_0 = h\nu_0 \Rightarrow \nu_0 = \frac{\phi_0}{h}$ 1

$\nu_0 = \frac{2.14 \times 1.6 \times 10^{-19}}{6.63 \times 10^{-34}}$ 1

$\boxed{\nu_0 = 5.16 \times 10^{14}\ \text{Hz}}$ 1

34. Give any three experimental results of photoelectric effect. [MQP-4, Q.35]

Ans: (i) Photoelectric effect is an instantaneous process. 1

(ii) For a given frequency of incident light, photoelectric current is directly proportional to the intensity of incident light. +

(iii) For any metal surface there exists a minimum frequency of incident light (or threshold frequency) below which there is no photoelectric emission. 1

(iv) Above threshold frequency, the maximum kinetic energy (or stopping potential) of photoelectrons is directly proportional to the frequency of incident light. +

(v) The maximum kinetic energy of photoelectrons is independent of the intensity of the incident light when the frequency is kept constant. 1

(vi) The photoelectric current is independent of the frequency of incident radiation. (Any three)

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Exam-1 March 2026, Q.35

35. Mention Einstein's photoelectric equation. Using the equation, explain any two experimental results of photoelectric effect. [MQP-5, Q.35]

Ans: Einstein's photoelectric equation: $K_{max} = h\nu - \phi_0$ OR $K_{max} = h\nu - h\nu_0$ OR $eV_0 = h\nu - \phi_0$, where K_{max} = maximum kinetic energy of photoelectrons, ν = frequency of incident light, h = Planck's constant, ϕ_0 = work function of the metal. 1

OFFICIAL ANSWER Equation: KSEAB Scheme of Evaluation, II PUC Annual Exam March 2020, Q.25

(i) K_{max} depends linearly on ν and is independent of intensity, because one photon is absorbed by one electron. 1

(ii) Since $K_{max} \geq 0$, emission is possible only if $h\nu \geq \phi_0$, i.e. $\nu \geq \nu_0 = \phi_0/h$. This explains the threshold frequency. 1

(iii) Intensity ∝ number of photons, so photocurrent ∝ intensity. (Equation + any two results)

CHAPTER 13
Nuclei

36. Differentiate between nuclear fission and nuclear fusion. [MQP-1, Q.36]

Ans:

Nuclear fission	**Nuclear fusion**
The process in which a heavy nucleus splits into two nuclei of comparable masses with release of energy is known as fission.	The process in which two lighter nuclei combine to form a single nucleus with the release of energy is known as fusion.
Fission can take place at room temperature.	Fusion takes place only at high temperature.
Energy released per nucleon (or per unit mass) of the reactant is less.	Energy released per nucleon (or per unit mass) of the reactant is more.
Energy released per reaction is more.	Energy released per reaction is less.
Can be controlled.	Cannot be controlled.

1+1+1

(Any three, or any other correct difference)

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Annual Exam March 2023, Q.29

37. Calculate the binding energy of an alpha (α) particle in MeV from the following data. Mass of α-particle = 4.002603 u; mass of neutron = 1.008665 u; mass of proton = 1.007825 u. [MQP-2, Q.36]

Ans: Mass defect of alpha particle: $\Delta m = Zm_p + (A - Z)m_n - M_\alpha$ 1

$\Delta m = (2 \times 1.007825) + (2 \times 1.008665) - 4.002603 = 0.030377$ u 1

Binding energy of alpha particle: $E_b = \Delta m \times 931.5\ \text{MeV} = 0.030377 \times 931.5 \approx \boxed{28.3\ \text{MeV}}$ 1

Full marks should be awarded if 931 MeV or 932 MeV is used while finding binding energy. (Method as in KSEAB Scheme of Evaluation, II PUC Supplementary Exam September 2020, Q.37, where the same data are used.)

38. Mention three features of nuclear force. [MQP-3, Q.36]

Ans: (i) Nuclear force is the strongest force. 1

(ii) Nuclear force is short ranged. 1

(iii) Nuclear force shows the property of saturation. 1

(iv) Nuclear force does not depend on the electric charge (charge independent). (Any three)

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Exam-2 2026, Q.36

39. Calculate the energy equivalent of one atomic mass unit (1 u) in joules. Given: Mass of carbon-12 atom is 1.992647×10^{-26} kg. [MQP-4, Q.36]

Ans: $1\text{ u} = \dfrac{\text{mass of } {}^{12}\text{C atom}}{12} = \dfrac{1.992647 \times 10^{-26}}{12} = 1.6605 \times 10^{-27}\text{ kg}$ 1

$E = mc^2 = 1.6605 \times 10^{-27} \times (2.9979 \times 10^8)^2$ 1

$\boxed{E = 1.4924 \times 10^{-10}\text{ J}}$ (≈ 931.5 MeV) 1

40. What is meant by radioactive decay? Name any two types of radioactive decay. [MQP-5, Q.36]

Ans: The spontaneous disintegration of an unstable nucleus by emitting α, β or γ radiation is called radioactive decay. 1

(i) α-decay (emission of a helium nucleus ${}^4_2\text{He}$) 1

(ii) β-decay (emission of an electron or positron) / (iii) γ-decay (emission of a high-energy photon) 1

(Any two types)

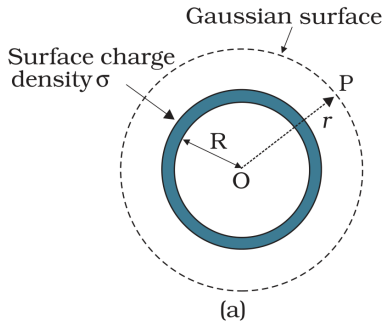
PART – D
FIVE MARKS QUESTIONS

MARKS

CHAPTER 1
Electric Charges and Fields

1. Using Gauss's law, arrive at the expression for electric field at a point outside a uniformly charged spherical shell. Also write the expression for the electric field at a point on the surface of the charged spherical shell. [MQP-2, Q.37]

Ans: **Statement:** The total electric flux (ϕ) through any closed surface is $1/\varepsilon_0$ times the net charge enclosed by the closed surface. 1



NCERT Fig. 1.28 (a) — Gaussian surface for a point with $r > R$ (Physics Part I, Ch. 1)

Consider a uniformly charged thin spherical shell of radius R. Let q be the charge distributed on the shell. It produces a spherically symmetric electric field. Let P be a point at distance r from the centre of the shell. To determine the field at P, imagine a Gaussian sphere of radius r with O as its centre, as shown in the figure. 1

The total electric flux through the Gaussian surface is: ϕ = electric intensity × area ($\because \vec{E} \parallel \vec{A}$), 1

$\phi = E \times 4\pi r^2$ .....(1)

According to Gauss's theorem: $\phi = \dfrac{q}{\varepsilon_0}$ .....(2) 1

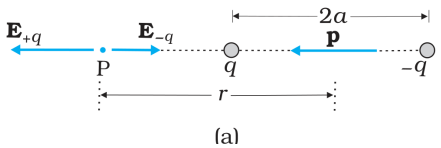
From (1) and (2): $E \times 4\pi r^2 = \dfrac{q}{\varepsilon_0} \Rightarrow \boxed{E = \dfrac{1}{4\pi\varepsilon_0}\dfrac{q}{r^2}}$

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Annual Exam March 2019, Q.27 (also II PUC Exam-2 April/May 2025, Q.37)

On the surface ($r = R$): $E = \dfrac{1}{4\pi\varepsilon_0}\dfrac{q}{R^2} = \dfrac{\sigma}{\varepsilon_0}$ $(q = 4\pi R^2\sigma)$ 1

2. Derive an expression for the electric field at a point on the axis of an electric dipole. [MQP-3, Q.37]

Ans: Consider a dipole of charges $-q$ at A and $+q$ at B separated by $2a$, with centre O. Let P be a point on the axis at distance r from O, on the side of $+q$. Let $\hat{p}$ be the unit vector along the dipole axis (from $-q$ to $+q$). 1



NCERT Fig. 1.17 (a) — Electric field of a dipole at a point on the axis (Physics Part I, Ch. 1)

Field at P due to $-q$ (distance $r+a$): $\vec{E}_{-q} = -\dfrac{q}{4\pi\varepsilon_0(r+a)^2}\,\hat{p}$ 1

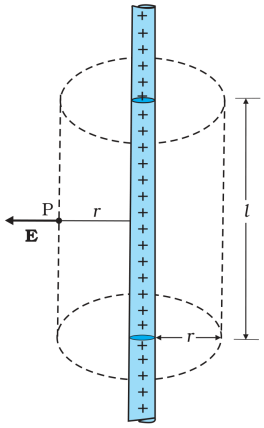
Field at P due to $+q$ (distance $r-a$): $\vec{E}_{+q} = \dfrac{q}{4\pi\varepsilon_0(r-a)^2}\,\hat{p}$ 1

Total field: $\vec{E} = \dfrac{q}{4\pi\varepsilon_0}\left[\dfrac{1}{(r-a)^2} - \dfrac{1}{(r+a)^2}\right]\hat{p} = \dfrac{q}{4\pi\varepsilon_0}\dfrac{4ar}{(r^2-a^2)^2}\,\hat{p}$ 1

For $r \gg a$: $\boxed{\vec{E} = \dfrac{2\vec{p}}{4\pi\varepsilon_0 r^3}}$, where $p = 2aq$ is the dipole moment. The field is along $\vec{p}$. 1

3. State Gauss law in electrostatics. Using it, derive the expression for the electric field at a point due to an infinitely long, straight, uniformly charged wire. [MQP-5, Q.37]

Ans: **Statement:** The electric flux through any closed surface is equal to $\dfrac{1}{\varepsilon_0}$ times the total (net) charge enclosed by the surface S. 1



(b)

NCERT Fig. 1.26 — Field of an infinitely long straight wire and its Gaussian surface (Physics Part I, Ch. 1)

Diagram. Consider an infinitely long uniformly charged straight wire with uniform linear charge density λ. Let us calculate the electric field at point P at a distance r from this wire. We will consider a cylindrical Gaussian surface of radius r and length l as shown in the figure. 1

By definition, electric flux through the Gaussian cylinder is $\phi_E = \int \vec{E}\cdot d\vec{A} = E \times 2\pi rl$ .....(1) 1

By Gauss's law in electrostatics, flux through the closed surface is $\phi_E = \dfrac{q}{\varepsilon_0}$ and charge enclosed 1

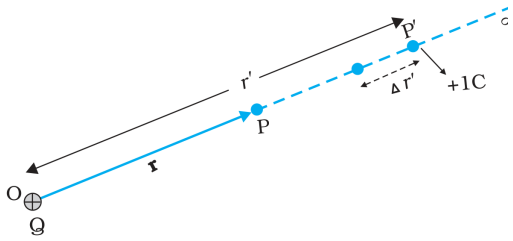
$q = \lambda l \Rightarrow \phi_E = \dfrac{\lambda l}{\varepsilon_0}$ .....(2)

Comparing (1) and (2): $E \times 2\pi rl = \dfrac{\lambda l}{\varepsilon_0} \Rightarrow \boxed{E = \dfrac{\lambda}{2\pi\varepsilon_0 r}}$ 1

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Exam-2 2026, Q.37

4. Derive an expression for electric potential at a point due to a point charge. [MQP-1, Q.37]

Ans: Let us calculate the potential due to a charge Q at a point P at a distance r from Q. Suppose a unit positive charge is brought radially from infinity to P. Consider an intermediate point A at a distance x from the charge. Let the unit positive charge be displaced by a small displacement dx. 1



NCERT Fig. 2.3 — Work done in bringing a unit positive test charge from infinity to the point P (Physics Part I, Ch. 2)

The force at this point is given by: $F = \dfrac{1}{4\pi\varepsilon_0}\dfrac{Q \times 1}{x^2}$ 1

The work done to move the unit positive charge through a distance dx is: 1

$dW = F\,dx\cos\theta = -\dfrac{1}{4\pi\varepsilon_0}\dfrac{Q}{x^2}\,dx$

Therefore, the total work done, which is equal to the potential at the point, is given by: 1

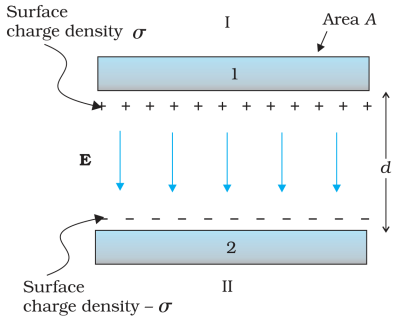
$$W = \int dW = \int_{\infty}^{r} -\frac{1}{4\pi\varepsilon_0}\frac{Q}{x^2}\,dx = -\frac{Q}{4\pi\varepsilon_0}\int_{\infty}^{r}\frac{1}{x^2}\,dx \Rightarrow W = \frac{Q}{4\pi\varepsilon_0}\left[\frac{1}{r} - \frac{1}{\infty}\right] = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r}$$

This is the work done to move the unit positive charge from infinity to point P. By definition, it is equal to the electric potential at P: $W = V, \therefore$ $\boxed{V = \dfrac{1}{4\pi\varepsilon_0}\dfrac{Q}{r}}$ 1

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Exam-1 March 2025, Q.37

5. Derive an expression for the capacitance of a parallel plate capacitor with air between the plates. Also write the general expression for the capacitance of a parallel plate capacitor with a dielectric medium. [MQP-4, Q.37]

Ans: The capacitor has two parallel plates, each of area A, separated by distance d (air in between). Plate 1 carries charge $+Q$ and plate 2 carries $-Q$. Surface charge density $\sigma = \dfrac{Q}{A}$. 1



NCERT Fig. 2.25 — The parallel plate capacitor (Physics Part I, Ch. 2)

Field between the plates (outside it is zero): $E = \dfrac{\sigma}{2\varepsilon_0} + \dfrac{\sigma}{2\varepsilon_0} = \dfrac{\sigma}{\varepsilon_0} = \dfrac{Q}{\varepsilon_0 A}$ 1

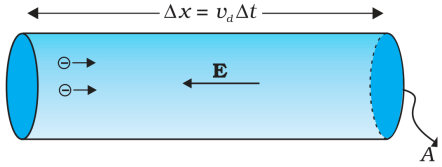
Potential difference between the plates: $V = Ed = \dfrac{Qd}{\varepsilon_0 A}$ 1

Capacitance: $C = \dfrac{Q}{V} \Rightarrow \boxed{C = \dfrac{\varepsilon_0 A}{d}}$ 1

With a dielectric of dielectric constant K between the plates: $\boxed{C = \dfrac{K\varepsilon_0 A}{d}}$ 1

CHAPTER 3
Current Electricity

6. Assuming the expression for drift velocity, derive the expression for conductivity of a material $\sigma = \dfrac{ne^2\tau}{m}$. [MQP-1, Q.38]



NCERT Fig. 3.4 — Current in a metallic conductor (Physics Part I, Ch. 3)

Ans: Diagram OR explanation as given below. Volume = $A(\Delta x) = A\, v_d(\Delta t)$. Let n be the number of free electrons per unit volume (number density) in the material; then there are $n\, v_d(\Delta t)A$ electrons in this volume. The electrons drift opposite to the electric field direction. (A = area of cross-section, v_d = drift velocity) 1

The amount of charge crossing the area A to the left in time Δt is $I(\Delta t) = neAv_d(\Delta t)$ OR current $I = neAv_d$ ..... (1) 1

Magnitude of drift velocity of electrons is $v_d = \left(\dfrac{e\tau}{m}\right)E$ ..... (2) 1

(e = magnitude of electron charge, τ = relaxation time, E = electric field, m = mass of electron)

Substituting v_d from (2) in (1): $I = neA\left(\dfrac{e\tau}{m}\right)E = \left(\dfrac{ne^2\tau A}{m}\right)E$. Thus, current density 1

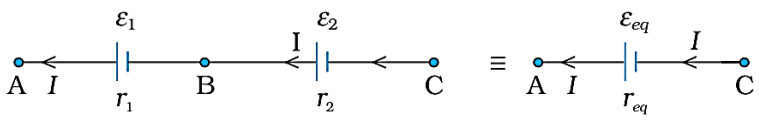
$j = \dfrac{I}{A} = \left(\dfrac{ne^2\tau}{m}\right)E$ ..... (3)

Also $j = \sigma E$ ..... (4). From (3) and (4), conductivity: $\boxed{\sigma = \dfrac{ne^2\tau}{m}}$ 1

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Annual Exam March 2023, Q.40

7. Obtain the expressions for equivalent emf and equivalent internal resistance of two cells of different emfs and different internal resistances connected in series. [MQP-2, Q.38]

Ans: Consider two cells ε_1, r_1 and ε_2, r_2 connected in series as shown in the figure. Let the equivalent cell be ε_{eq}, r_{eq}. Let the terminal potential differences for the first cell be V_1, for the second cell V_2 and for the combination V.



Two cells in series and the equivalent cell (from the KSEAB scheme)

1

For the first cell, $V_1 = \varepsilon_1 - Ir_1$ and for the second cell, $V_2 = \varepsilon_2 - Ir_2$. 1

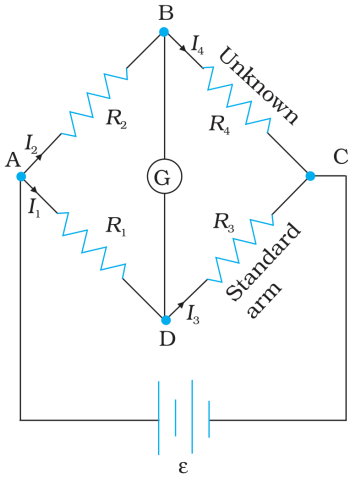
As the cells are in series, $V = V_1 + V_2 \Rightarrow V = (\varepsilon_1 - Ir_1) + (\varepsilon_2 - Ir_2) = (\varepsilon_1 + \varepsilon_2) - I(r_1 + r_2)$. 1

For the equivalent cell, $V = \varepsilon_{eq} - Ir_{eq}$. 1

By comparing the last two equations: $\boxed{\varepsilon_{eq} = \varepsilon_1 + \varepsilon_2}$ and $\boxed{r_{eq} = r_1 + r_2}$ 1

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Exam-1 March 2026, Q.38

8. Using Kirchhoff's laws, arrive at the balancing condition of Wheatstone network. [MQP-3, Q.38]



NCERT Fig. 3.18 — Wheatstone bridge (Physics Part I, Ch. 3)

Ans: Labelled circuit diagram with arrows for currents. Let four resistors R_1, R_2, R_3 and R_4 be connected to form a Wheatstone bridge. Let the currents through the resistors be I_1, I_2, I_3 and I_4 respectively. When the bridge is balanced, no current flows through the galvanometer G, i.e. $I_g = 0$. 1

Applying Kirchhoff's junction rule to the junctions D and B when $I_g = 0$ gives $I_1 = I_3$ and $I_2 = I_4$. 1

Taking $I_g = 0$ and p.d. across G is 0:

Applying Kirchhoff's loop rule to loop ABDA gives: $-I_2R_2 + 0 + I_1R_1 = 0$ .....(1)

Applying Kirchhoff's loop rule to loop BCDB gives: $-I_4R_4 + I_3R_3 + 0 = 0$. Using $I_3 = I_1$, $I_4 = I_2$ 1

we get $I_2R_4 - I_1R_3 = 0$ ..... (2)

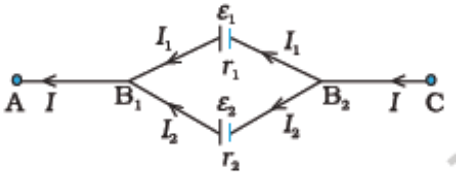
From (1): $I_2R_2 = I_1R_1 \Rightarrow \dfrac{I_1}{I_2} = \dfrac{R_2}{R_1}$ .....(3); From (2): $I_2R_4 = I_1R_3 \Rightarrow \dfrac{I_1}{I_2} = \dfrac{R_4}{R_3}$ .....(4) 1

From (3) and (4), the balance condition is $\boxed{\dfrac{R_2}{R_1} = \dfrac{R_4}{R_3}}$ 1

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Exam-3 2025, Q.38

9. Obtain the expressions for equivalent emf and equivalent internal resistance of two cells of different emfs and different internal resistances connected in parallel. [MQP-4, Q.38]

Ans: Let us consider the parallel combination of two cells with positive terminals connected to B_1 and negative terminals connected to B_2. E_1, E_2 are the emfs of the two cells and r_1, r_2 their internal resistances. I_1 and I_2 are the currents leaving the positive electrodes of the cells.



Two cells in parallel (from the KSEAB scheme)

Since as much charge flows in as out, we have $I = I_1 + I_2$. Let $V(B_1)$ and $V(B_2)$ be the potentials at B_1 and B_2. 1

Considering the first cell: $V = V(B_1) - V(B_2) = E_1 - I_1 r_1 \Rightarrow I_1 = \dfrac{E_1 - V}{r_1}$. Similarly for the second cell: $V = E_2 - I_2 r_2 \Rightarrow I_2 = \dfrac{E_2 - V}{r_2}$ 1

Therefore $I = \dfrac{E_1 - V}{r_1} + \dfrac{E_2 - V}{r_2} = \dfrac{E_1}{r_1} + \dfrac{E_2}{r_2} - V\left(\dfrac{1}{r_1} + \dfrac{1}{r_2}\right) \Rightarrow V = \left(\dfrac{E_1 r_2 + E_2 r_1}{r_1 + r_2}\right) - I\left(\dfrac{r_1 r_2}{r_1 + r_2}\right)$ 1

(1)

If the combination of cells is replaced by a single cell of emf E_{eq} and internal resistance r_{eq}, then 1

$V = E_{eq} - I r_{eq}$ ..... (2)

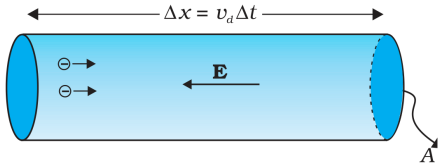
From (1) and (2): $\boxed{E_{eq} = \dfrac{E_1 r_2 + E_2 r_1}{r_1 + r_2}}$ and $\boxed{r_{eq} = \dfrac{r_1 r_2}{r_1 + r_2}}$ 1

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Supplementary Exam May 2023, Q.40

10. What is meant by drift velocity of free electrons in a conductor? Derive an expression for electric current in terms of drift velocity and number density of free electrons. [MQP-5, Q.38]

Ans: **Drift velocity:** The average velocity gained by the free electrons of a conductor in the opposite direction (against) to the applied electric field. 1

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Supplementary Exam September 2020, Q.12



NCERT Fig. 3.4 — Current in a metallic conductor (Physics Part I, Ch. 3)

Consider a conductor of cross-sectional area A with n free electrons per unit volume, each of charge e, drifting with speed v_d. 1

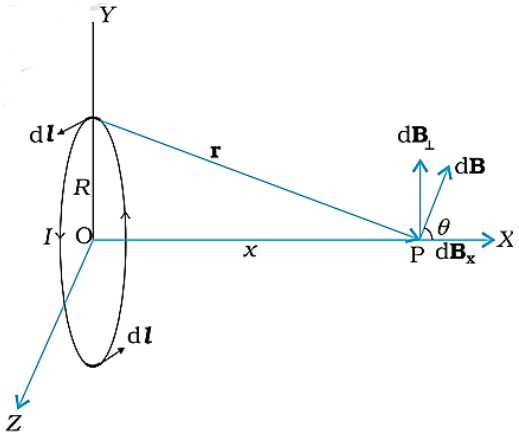
In time Δt, all electrons within a length $v_d \Delta t$ cross the area A. Volume = $A v_d \Delta t$, so the number of electrons = $n A v_d \Delta t$. 1

Charge crossing in time Δt: $\Delta q = n e A v_d \Delta t$ 1

Current: $I = \dfrac{\Delta q}{\Delta t} \Rightarrow \boxed{I = n e A v_d}$ 1

11. Derive an expression for the magnetic field at a point on the axis of a circular current loop using Biot–Savart law. [MQP-1, Q.39]

Ans: Consider a circular coil of radius R carrying a current I placed in the YZ plane. Let us calculate the magnetic field at a point P along the axis of the coil. The distance of the point P from the centre of the circular coil is x.



Circular current loop and the field at an axial point P (from the KSEAB scheme)

1

The magnitude of the magnetic field at P due to a small element of the coil is: 1

$$dB = \frac{\mu_0}{4\pi}\frac{I|d\vec{l}\times\vec{r}|}{r^3} = \frac{\mu_0}{4\pi}\frac{I\,dl}{r^2}$$

When we add $d\vec{B}$ to get the total magnetic field at P, the vertical components of $d\vec{B}$ cancel and the horizontal components (along the x-axis) get added up. The x-component of $d\vec{B}$ is: 1

$$dB_x = \frac{\mu_0}{4\pi}\frac{I\,dl}{r^2}\cos\theta$$

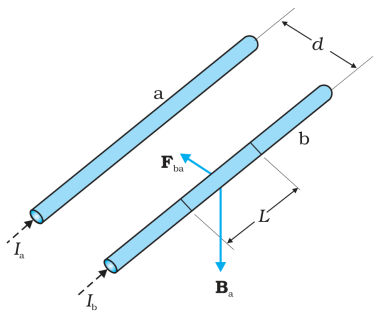
The total magnetic field at P is: $B = \sum dB_x = \sum \frac{\mu_0}{4\pi}\frac{I\,dl}{r^2}\cos\theta$ 1

But $\sum dl = 2\pi R$, $\cos\theta = \frac{R}{r}$ and $r = (R^2 + x^2)^{1/2}$. Substituting: $\boxed{B = \frac{\mu_0 I R^2}{2(R^2+x^2)^{3/2}}}$ 1

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Exam-1 March 2026, Q.39

12. Derive the expression for force per unit length on two infinitely long straight parallel conductors carrying currents. Hence define 'ampere'. [MQP-2, Q.39]

Ans: Labelled diagram with currents, field and force ($\vec{F}_{ab}$ and $\vec{B}_b$ are not necessary in the diagram). 1



NCERT Fig. 4.17 — Two long straight parallel conductors carrying steady currents I_a and I_b (Physics Part I, Ch. 4)

The magnetic field at the location of 'b' produced by the conductor 'a' is $B_a = \dfrac{\mu_0 I_a}{2\pi d}$ ..... (1) 1

The magnetic force on a segment L of the conductor 'b' due to 'a' is $F_{ba} = B_a I_b L \sin 90°$. Using (1): 1

$F_{ba} = \dfrac{\mu_0 I_a I_b L}{2\pi d}$. This force F_{ba} is towards conductor 'a'.

Force per unit length on the conductor 'b' is $\dfrac{F_{ba}}{L} = \dfrac{\mu_0}{4\pi}\left(\dfrac{2I_a I_b}{d}\right) = \dfrac{\mu_0 I_a I_b}{2\pi d}$. Similarly, force per 1

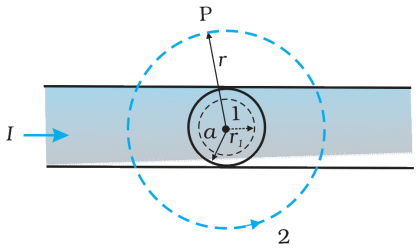
unit length on the conductor 'a' is $\dfrac{F_{ab}}{L} = \dfrac{\mu_0 I_a I_b}{2\pi d}$ and $\vec{F}_{ab} = -\vec{F}_{ba}$.

Definition of ampere: One ampere is that steady current which, when maintained in each of two infinitely long, straight, parallel conductors of negligible cross-section and placed one metre apart in vacuum (or free space or air), would produce a force of 2×10^{-7} newton per metre of length on each other. 1

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Exam-2 2026, Q.39

13. (i) State Ampere's circuital law. (1) (ii) Using Ampere's circuital law, derive the expression for the magnetic field at a point due to an infinitely long current carrying straight wire. (4) [MQP-3, Q.39]

Ans: (i) **Ampere's circuital law:** The line integral of $\vec{B} \cdot d\vec{l}$ over the closed loop is equal to μ_0 times the total current passing through the surface. OR $\oint \vec{B} \cdot d\vec{l} = \mu_0 I$, where I – current, B – magnetic field and dl – elemental length. 1



NCERT Fig. 4.13 — Ampere's circuital law for an infinitely long straight wire (Physics Part I, Ch. 4)

(ii) **Derivation:** Let P be a point at a distance r from the straight wire. Consider a circle of radius r around the wire passing through P as an Amperian loop. The magnetic field B at P due to the current in the wire is along the tangent to this circle. Hence the magnetic field B is parallel to the line element dl at P. 1

From Ampere's circuital law, $\oint \vec{B} \cdot d\vec{l} = \mu_0 I$ 1

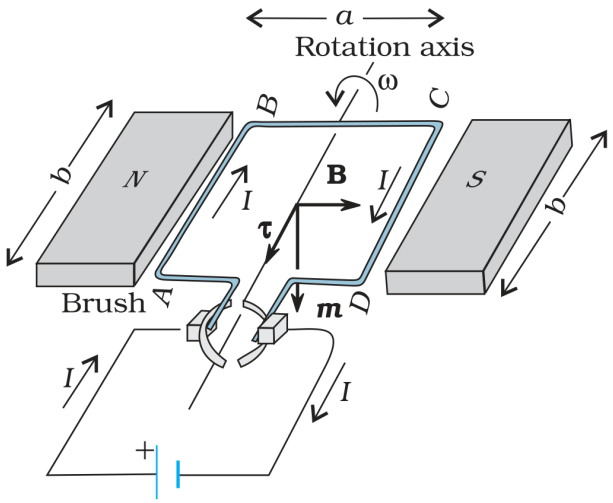
$\oint \vec{B} \cdot d\vec{l} = \oint B\, dl \cos 0° = B \times 2\pi r$. Thus $B \times 2\pi r = \mu_0 I$ 1

The magnetic field at a distance r outside the wire is $\boxed{B = \dfrac{\mu_0 I}{2\pi r}}$ 1

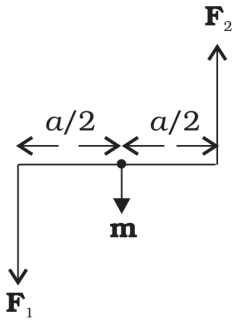
OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Supplementary Exam August 2021, Q.23

14. (i) Obtain the expression for the torque acting on a rectangular current loop placed in a uniform magnetic field. (3) (ii) When is the torque on the current loop (a) maximum and (b) zero? (2) [MQP-4, Q.39]

Ans: (i) A rectangular loop ABCD with sides a and b carries current I in a uniform field $\vec{B}$; its normal (along $\vec{m}$) makes angle θ with $\vec{B}$. The forces on arms BC and DA are equal, opposite and collinear, so they cancel. The forces on arms AB and CD are $F_1 = F_2 = IbB$; they are equal, opposite and not collinear, so they form a couple. 1



(a)



(b)

NCERT Fig. 4.18 — Rectangular current-carrying coil in a uniform magnetic field (Physics Part I, Ch. 4)

Torque: $\tau = F_1 \frac{a}{2} \sin\theta + F_2 \frac{a}{2} \sin\theta = IabB \sin\theta$ 1

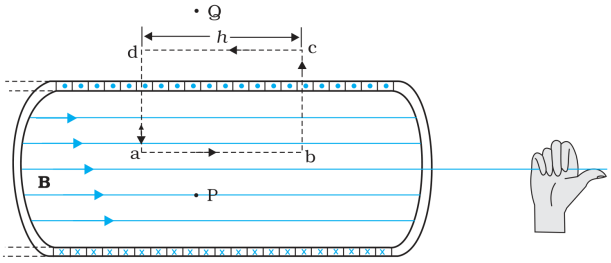
Since $A = ab$ and $m = IA$: $\boxed{\tau = mB \sin\theta}$, or $\vec{\tau} = \vec{m} \times \vec{B}$ (for N turns, $m = NIA$) 1

(ii) (a) **Maximum** ($\tau = mB$) when $\theta = 90°$, i.e. the plane of the loop is parallel to $\vec{B}$. 1

(b) **Zero** when $\theta = 0°$ or $180°$, i.e. the plane of the loop is perpendicular to $\vec{B}$. 1

15. Derive the expression for the magnetic field at a point inside an air cored long current carrying solenoid using Ampere's circuital law. [MQP-5, Q.39]

Ans: Consider a long solenoid with n turns per unit length carrying current I. Inside it the field is uniform and along the axis; outside it is nearly zero. Take a rectangular Amperian loop abcd with side ab ($= h$) inside, parallel to the axis. 1



NCERT Fig. 4.16 — The magnetic field of a very long solenoid; Amperian loop abcd (Physics Part I, Ch. 4)

Along the perpendicular sides bc and da, $\vec{B} \perp d\vec{l}$ (and $B = 0$ outside), so they contribute nothing. 1

Along cd (outside), $B = 0$.

$\oint \vec{B} \cdot d\vec{l} = \int_a^b \vec{B} \cdot d\vec{l} = Bh$ 1

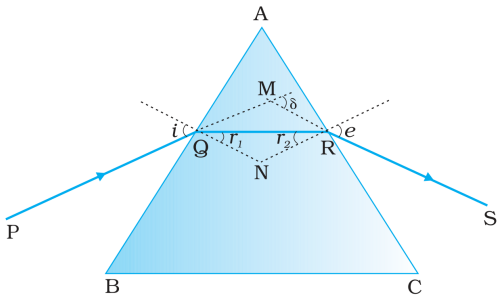
Current enclosed by the loop: $I_e = I(nh)$ 1

By Ampere's law, $Bh = \mu_0(nh)I \Rightarrow \boxed{B = \mu_0 nI}$ 1

CHAPTER 9

Ray Optics and Optical Instruments

16. Derive the expression for refractive index of the material of a prism in terms of angle of the prism and angle of minimum deviation. [MQP-3, Q.40]



NCERT Fig. 9.21 — A ray of light passing through a triangular glass prism (Physics Part II, Ch. 9)

Ans: Labelled ray diagram. In the quadrilateral AQNR, two of the angles (at the vertices Q and R) are right angles. Therefore the sum of the other angles of the quadrilateral is 180°: 1+1

$\angle A + \angle QNR = 180°$. In $\triangle QNR$: $r_1 + r_2 + \angle QNR = 180°$. Comparing: $A = r_1 + r_2$ .....(1)

The total deviation δ is the sum of deviations at the two faces: $\delta = (i - r_1) + (e - r_2)$; using 1

$A = r_1 + r_2$: $\delta = i + e - A$ .....(2)

At the minimum deviation position, $\delta = D_m$, $i = e$ and $r_1 = r_2$. Eq. (1) becomes 1

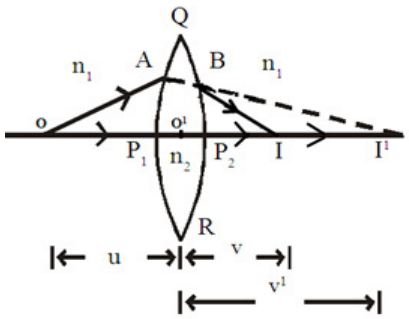
$A = 2r_1 \Rightarrow r_1 = A/2$ and Eq. (2) becomes $D_m = 2i - A$ or $i = (A + D_m)/2$.

From Snell's law, refractive index: $\boxed{n_{21} = \frac{n_2}{n_1} = \frac{\sin i}{\sin r_1} = \frac{\sin\left(\frac{A+D_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}}$ 1

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Annual Exam March 2023, Q.42

17. Derive Lens maker's formula for a convex lens. [MQP-4, Q.40]

Ans: Let O be a point object placed on the principal axis of a thin convex lens of focal length f. n_1 is the refractive index of the medium in which the object is present and n_2 that of the material of the lens.



Refraction through a thin convex lens (from the KSEAB scheme)

1

(i) For the refraction at the surface QP_1R of radius of curvature R_1: a real image I_1 is formed in the medium of refractive index n_2. For the refraction $\frac{n_1}{-u} + \frac{n_2}{v_1} = \frac{n_2 - n_1}{R_1}$ .....(1) 1

(ii) For the refraction at the surface QP_2R of radius of curvature R_2: the final image is formed at I, and I_1 acts as a virtual object. For the refraction $\frac{n_2}{-v_1} + \frac{n_1}{v} = \frac{n_2 - n_1}{-R_2}$ ..... (2) 1

Equation (1) + (2) gives $\frac{n_1}{-u} + \frac{n_1}{v} = \frac{n_2 - n_1}{R_1} - \frac{n_2 - n_1}{R_2} \Rightarrow \left(\frac{1}{-u} + \frac{1}{v}\right) = \left(\frac{n_2}{n_1} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$ 1

If $u = \infty$ then $v = f$. Therefore $\boxed{\frac{1}{f} = \left(\frac{n_2}{n_1} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)}$ OR $\frac{1}{f} = (n_{21} - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$. This is the lens maker's formula. 1

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Supplementary Exam May 2023, Q.42

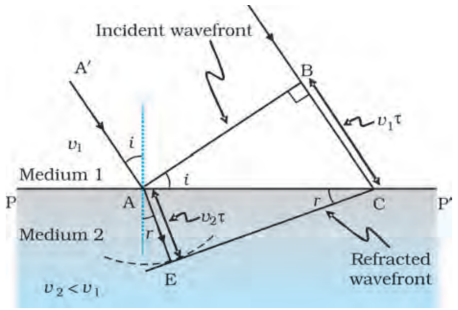
CHAPTER 10
Wave Optics

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18. (i) What is a wavefront? (1) (ii) Using Huygens principle, derive Snell's law for a plane wave undergoing refraction. (4) [MQP-1, Q.40]

Ans: (i) The locus of points which oscillate in the same phase is called a wavefront. OR A wavefront is defined as a surface of constant phase. 1

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Supplementary Exam August 2021, Q.18



NCERT Fig. 10.4 — Refraction of a plane wave AB incident at angle i on the surface PP′ (Physics Part II, Ch. 10)

(ii) Consider a plane wavefront passing from medium 1 (speed v_1) to medium 2 (speed v_2) as shown in the figure. Let AB be the incident wavefront and EC the refracted wavefront. Let τ be the time taken by the wavefront to travel from B to C, so $BC = v_1\tau$; in this time the secondary wavelet from A travels $AE = v_2\tau$. 1

In triangle ABC: $\sin i = \dfrac{BC}{AC} = \dfrac{v_1\tau}{AC}$ ..... (1) 1

In triangle AEC: $\sin r = \dfrac{AE}{AC} = \dfrac{v_2\tau}{AC}$ ..... (2) 1

From (1) and (2): $\dfrac{\sin i}{\sin r} = \dfrac{v_1\tau}{v_2\tau} = \dfrac{v_1}{v_2}$ = constant. Since $n_1 = c/v_1$ and $n_2 = c/v_2$: 1

$\boxed{n_1 \sin i = n_2 \sin r}$ — Snell's law.

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Exam-2 April/May 2025, Q.40(a)

19. (i) Two coherent waves of same amplitude and constant phase difference undergo interference. Obtain an expression for resultant displacement. (3) (ii) Mention two uses of polaroids. (2) [MQP-2, Q.40]

Ans: (i) Let the displacements due to the two coherent waves at a point be $y_1 = a\cos\omega t$ and 1

$y_2 = a\cos(\omega t + \phi)$, where ϕ is the constant phase difference.

By the superposition principle: $y = y_1 + y_2 = a[\cos\omega t + \cos(\omega t + \phi)]$. Using 1

$\cos A + \cos B = 2\cos\dfrac{A+B}{2}\cos\dfrac{A-B}{2}$:

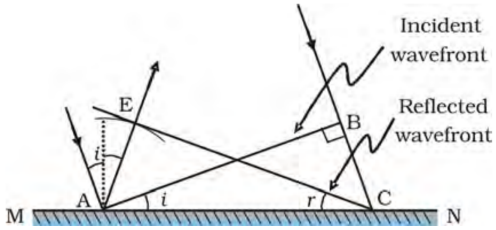
$\boxed{y = 2a\cos\left(\dfrac{\phi}{2}\right)\cos\left(\omega t + \dfrac{\phi}{2}\right)}$. The resultant amplitude is $2a\cos(\phi/2)$, and the intensity is 1

$I = 4I_0\cos^2(\phi/2)$.

(ii) Uses of polaroids: sunglasses, windowpanes, photographic cameras, 3D movie cameras. (ANY TWO — any other correct use) 1+1

OFFICIAL ANSWER Part (ii): KSEAB Scheme of Evaluation, II PUC Exam-2 2026, Q.40(ii)

20. (i) Using Huygens principle, show that the angle of incidence is equal to the angle of reflection during reflection of a plane wavefront at a surface. (3) (ii) Write the conditions for constructive interference and destructive interference in terms of path difference. (2) [MQP-5, Q.40]



NCERT Fig. 10.6 — Reflection of a plane wave AB by the reflecting surface MN (Physics Part II, Ch. 10)

Ans: (i) Labelled diagram. MN → reflecting plane surface (mirror); i → angle of incidence; r → angle of reflection; v → speed of light wave. 1

Let τ be the time taken by the wavefront to travel from the point B to C; then distance $BC = v\tau$. In order to construct the reflected wavefront, a sphere of radius $v\tau$ is drawn from the point A as shown in the figure. Let CE represent the tangent plane drawn from the point C to this sphere. 1

Then $AE = BC = v\tau$, $\angle ABC = \angle CEA = 90°$ and AC is common.

The triangles ABC and CEA are congruent. Thus, $\boxed{i = r}$ 1

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Exam-2 2026, Q.40(i)

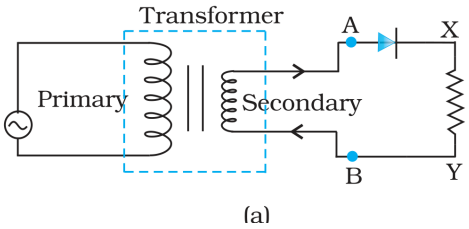
(ii) Condition for constructive interference: path difference $\delta = n\lambda$ 1

Condition for destructive interference: path difference $\delta = (2n+1)\dfrac{\lambda}{2}$ or $\left(n + \dfrac{1}{2}\right)\lambda$, where n is the order of the fringe and λ is the wavelength of the light. 1

OFFICIAL ANSWER Part (ii): KSEAB Scheme of Evaluation, II PUC Exam-2 March 2024, Q.42(b)

21. What is a half-wave rectifier? Draw a neat, labelled circuit diagram and input–output waveforms for half-wave rectifier. Explain the working of a half-wave rectifier. [MQP-1, Q.41]

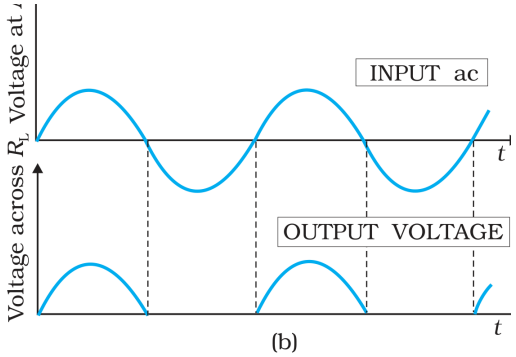
Ans: A half-wave rectifier is a circuit (device) that rectifies only one half cycle of AC (converts AC into DC during one half cycle). 1



(a)

NCERT Fig. 14.18 (a) — Half-wave rectifier circuit (Physics Part II, Ch. 14)

Circuit diagram (as above) 1



(b)

NCERT Fig. 14.18 (b) — Input ac voltage and output voltage waveforms (Physics Part II, Ch. 14)

Waveform (as above) 1

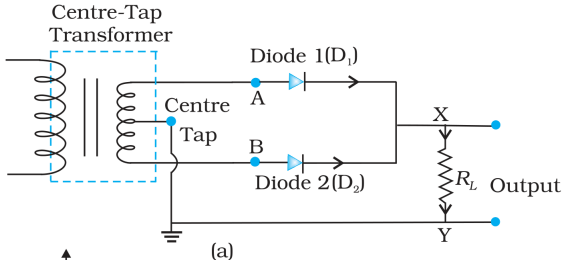
Working: During the positive half cycle of AC, A is positive and B is negative. The diode becomes forward biased and it conducts. The output appears across the resistance R_L. 1

During the negative half cycle of AC, A becomes negative and B becomes positive. The diode is reverse biased, and it does not conduct. No output appears in this case. 1

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Exam-2 2026, Q.41 and Exam-1 March 2026, Q.41

22. What is meant by rectification? Explain the working of a p-n junction diode as a full-wave rectifier, with the help of relevant circuit diagram and input–output waveforms. [MQP-2, Q.41]

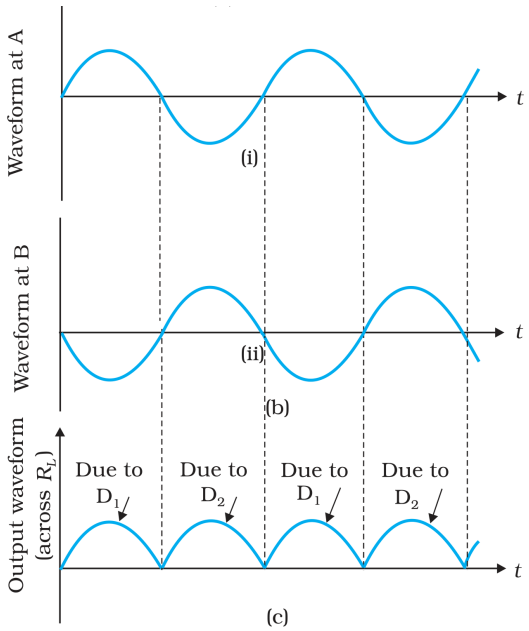
Ans: (a) The process of conversion of AC into DC is called rectification. 1



(a)

NCERT Fig. 14.19 (a) — A full-wave rectifier circuit (Physics Part II, Ch. 14)

(b) Circuit diagram (as above) 1



NCERT Fig. 14.19 (b), (c) — Input waveforms at A and B; output waveform across R_L (Physics Part II, Ch. 14)

Waveform (as above) 1

(c) During the positive half cycle of AC input, the diode D_1 is forward biased and conducts, while D_2 is reverse biased and does not conduct. So the output current flows through R_L as shown in the figure. 1

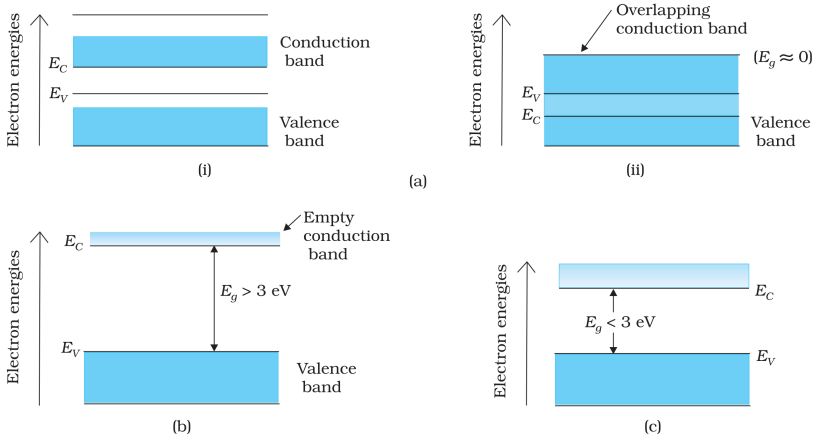
During the negative half cycle of AC input, D_2 is forward biased and conducts while D_1 is reverse biased and does not conduct. Again the current flows through R_L in the same direction. Thus there is current flow through R_L over the complete cycle of AC input in the same direction. 1

OFFICIAL ANSWER KSEAB Scheme of Evaluation, II PUC Annual Exam March 2023, Q.44

23. (i) Explain the formation of energy bands in solids. (2) (ii) On the basis of energy bands, distinguish between a conductor, a semiconductor and an insulator. (3) [MQP-3, Q.41]

Ans: (i) In an isolated atom, electrons occupy discrete energy levels. In a crystal, atoms are very close, so the outer electrons interact with neighbouring atoms. Each energy level then splits into a large number of closely spaced levels, which form an energy band. 1

The band containing the valence electrons is the valence band; the next higher band is the conduction band. They are separated by the forbidden energy gap E_g. 1



NCERT Fig. 14.2 — Energy bands of (a) metals, (b) insulators and (c) semiconductors (Physics Part II, Ch. 14)

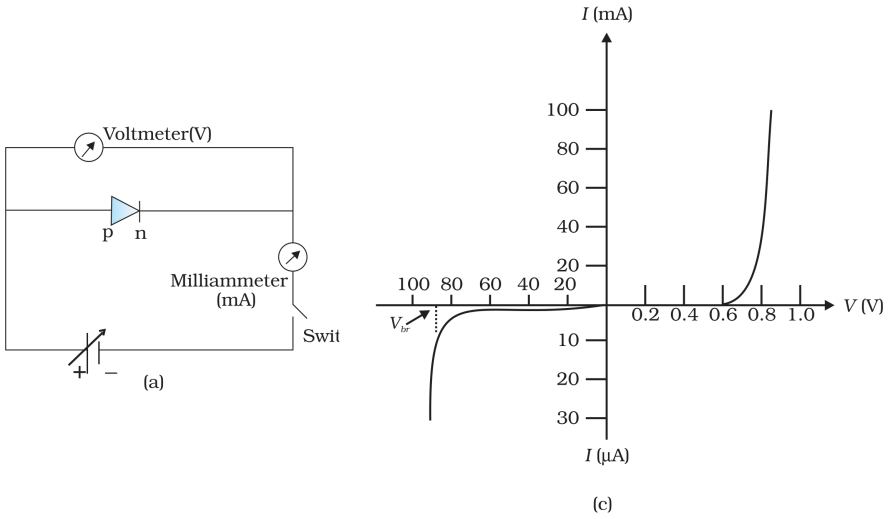
(ii) Solids are classified as conductors, semiconductors and insulators using energy band theory. In **conductors** the valence band and conduction band are overlapped. 1

In **semiconductors** there is a small energy gap (< 3 eV) between the valence band and the conduction band. 1

In **insulators** there is a large energy gap (> 3 eV) between the valence band and the conduction band. 1

OFFICIAL ANSWER Part (ii): KSEAB Scheme of Evaluation, II PUC Exam-2 April/May 2025, Q.41(a)

24. With the help of experimental circuit and V–I characteristics, explain the action of a p-n junction diode under forward bias. [MQP-4, Q.41]



NCERT Fig. 14.16 (a) Experimental circuit – diode in forward bias; (c) Typical V–I characteristics of a silicon diode (Physics Part II, Ch. 14)

1+1

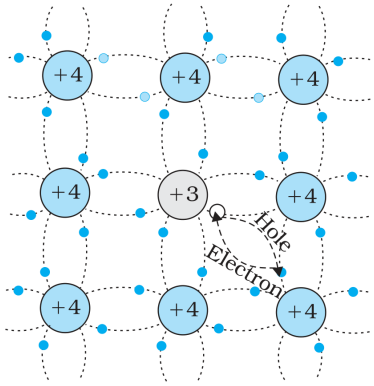
Ans: In forward bias, the p-side is connected to the positive terminal and the n-side to the negative terminal of the battery. The applied voltage opposes the barrier potential, so the effective barrier height and the width of the depletion layer decrease. 1

Initially the current is very small (negligible) until the applied voltage exceeds the threshold (cut-in/knee) voltage — about 0.7 V for Si and 0.2 V for Ge. 1

Beyond the threshold voltage, the diode current increases significantly (exponentially), of the order of mA, even for a small increase in voltage. 1

25. (i) With the help of a figure, explain how a p-type semiconductor is obtained from an intrinsic semiconductor. (3) (ii) What happens to the width of depletion layer of a p-n junction diode when it is (a) forward biased and (b) reverse biased? (2) [MQP-5, Q.41]

Ans: (i) When a trivalent impurity (B, Al, In) is added to pure Si or Ge, the impurity atom forms covalent bonds with only three neighbouring Si atoms. One bond remains incomplete, which creates a vacancy called a hole. 1



(a)

NCERT Fig. 14.8 (a) — Trivalent acceptor atom doped in a tetravalent Si or Ge lattice (Physics Part II, Ch. 14)

1

An electron from a neighbouring bond can jump into this hole, so holes become the majority carriers. The impurity atom accepts an electron, so it is called an acceptor, and the material is a p-type semiconductor. 1

(ii) (a) In forward bias, the width of the depletion layer **decreases**. 1

(b) In reverse bias, the width of the depletion layer **increases**. 1